

Global Cybersecurity Incident & Threat Analytics

Exploratory Data Analysis and Interactive Dashboard Development using Python and Power BI

1. Project Overview and Objective

Project Overview

The **Global Cybersecurity Incident and Threat Analytics** is a data analysis project that tracks and visualises over **3,400+ cyber incidents** across the globe. It analyses attack patterns, threat actor behaviours, and operational impacts through three core views:

- **Global Incident Overview:** Tracks incident growth over the years and highlights heavily targeted sectors such as Government, Healthcare, and Finance.
- **Threat and Attacker Analysis:** Maps threat actor origins and attack methodologies aligned with the **MITRE ATT&CK** framework, highlighting vectors like Phishing and Public Application Exploits.
- **Impact and Risk Analysis:** Evaluates total economic losses (**\$2.9B+**), system downtime duration, and incident severity distributions.

Project Objectives

- Transform complex, raw cybersecurity telemetry into intuitive, interactive reporting dashboards.
 - Identify prominent attack vectors to help organisations strengthen and prioritise their security **defences**.
 - Track nation-state threat actors and evaluate their active campaign patterns.
 - Quantify the financial and operational impact of cyber incidents to support risk assessment and decision-making.
-

2. Data Sources

- **Source Description and Timeline:**

European Repository of Cyber Incidents (EuRepoC) — Zenodo Open Research Repository EuRepoC Global Dataset1.3

<https://zenodo.org/records/14965395>

Data collected during 2000 to 2024

- **Domain:** Information Technology – Cybersecurity
-

3. Problem Statement

Cyberattacks are growing rapidly worldwide, but organisations struggle to defend themselves due to the following key challenges:

- **Unclear Threat Patterns:** With incidents spread across different years and countries, it is difficult to track how attack types and trends are shifting over time.
- **Limited Visibility into Attacker Methods:** Security teams often lack clear insight into threat actors, attribution sources, and common **MITRE ATT&CK** techniques used to breach systems.
- **Tracking Zero-Day Threats:** It is challenging to measure how often unknown vulnerabilities (zero-days) are exploited compared to standard attack types.
- **Measuring Real-World Impact:** Organisations struggle to evaluate the true severity of attacks, including operational downtime, intelligence leaks, and financial losses.
- **Lack of Interactive Tools:** Raw incident data is rarely presented in simple, interactive visual formats (like Power BI or Python dashboards) for quick analysis and decision-making.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
incident_id	Integer	Unique identifier assigned to each cybersecurity incident.
Name	String (Text)	Name or title associated with the cybersecurity incident.
start_date	Date	Date on which the cybersecurity incident started.
end_date	Date	Date on which the cybersecurity incident ended or was reported as ended.
incident_type	Categorical	Classification of the cybersecurity incident, such as disruption, data theft, ransomware, or hijacking.
receiver_country	Categorical	Country affected or targeted by the cybersecurity incident.
receiver_regions	Categorical	Geographical region associated with the affected target.

receiver_category	Categorical	Broad category of the organization or entity affected by the incident.
receiver_subcategory	Categorical	More specific classification of the affected organization or entity.
initiator_name	String (Text)	Name of the suspected or identified actor responsible for initiating the incident.
initiator_country	Categorical	Country associated with the suspected or identified initiator.
initiator_category	Categorical	Broad category of the actor or entity responsible for the cyber incident.
initiator_subcategory	Categorical	More specific classification of the initiator or threat actor.
attribution_type	Categorical	Type or level of attribution associated with the identified threat actor.
attributing_actor	String (Text)	Actor or entity to which responsibility for the incident is attributed.
cyber_conflict_issue	Categorical	Cyber conflict issue or context associated with the incident.
zero_days	Boolean / Categorical	Indicates whether the incident involved exploitation of a zero-day vulnerability.
mitre_initial_access	Categorical	MITRE ATT&CK technique used by the attacker to gain initial access.
mitre_impact	Categorical	MITRE ATT&CK technique associated with the impact caused by the incident.
has_disruption	Boolean	Indicates whether the incident resulted in disruption of systems, services, or operations.
data_theft	Boolean / Categorical	Indicates whether the incident involved theft or unauthorized extraction of data.
Hijacking	Boolean / Categorical	Indicates whether the incident involved hijacking or unauthorized takeover of systems, accounts, or resources.
impact_indicator_score	Integer	Score representing the overall impact or severity of the incident.

functional_impact	Categorical	Effect of the incident on the normal functioning or operations of the affected entity.
intelligence_impact	Categorical	Impact of the incident on information, intelligence, or sensitive knowledge.
economic_impact_value	Numeric (Integer)	Estimated monetary or economic value of the impact caused by the incident.
response_indicator	Categorical	Indicates the response or actions taken following the cybersecurity incident.
start_year	Integer	Year extracted from the incident start date for time-based analysis.

5. Tools & Technologies

Tool/Technology	Purpose
Python	Data cleaning, preprocessing, Exploratory Data Analysis (EDA), and data visualization.
Pandas	Data manipulation, transformation, handling missing values, and data analysis.
NumPy	Numerical operations and data processing.
Matplotlib	Creating charts and visualizations for data analysis.
Seaborn	Statistical data visualization and exploratory analysis.
DAX	Creating calculated measures and analytical metrics in Power BI.
Power BI	Creating interactive dashboards, reports, and data visualizations.

6. Data Pre-Processing (Python)

Tasks Performed:

- **Data Cleaning:** Checked and handled missing, inconsistent, and unavailable values in the dataset.
- **Data Transformation:** Standardized country names, categories, dates, and other relevant fields for consistent analysis.
- **Duplicate & Consistency Checking:** Checked the dataset for duplicate records and inconsistent entries.
- **Column Selection:** Selected 27 relevant columns from the original dataset for further analysis.

- **Date Processing:** Converted incident start and end dates into appropriate date formats and handled invalid or unavailable date values.
- **Categorical Data Cleaning:** Cleaned and standardized incident types, attribution types, countries, MITRE techniques, and impact-related categories.
- **Data Preparation:** Prepared the cleaned dataset for Exploratory Data Analysis in Python and interactive visualization in Power BI

```
df_selected["start_date"] = pd.to_datetime(df_selected["start_date"],format="%d.%m.%Y",errors="coerce")

df_selected["end_date"] = pd.to_datetime(df_selected["end_date"],format="%d.%m.%Y",errors="coerce")

df_selected[["start_date", "end_date"]].dtypes

0
start_date  datetime64[ns]
end_date    datetime64[ns]

dtype: object
```

```
def clean_multi_value(value):
    if value == "Not available":
        return value

    values = value.split(";")
    unique_values = []

    for item in values:
        item = item.strip()

        if item not in unique_values:
            unique_values.append(item)

    return ", ".join(unique_values)

df_selected["incident_type"] = df_selected["incident_type"].apply(clean_multi_value)

df_selected["incident_type"] = (
    df_selected["incident_type"]
    .str.replace("Not available, ", "", regex=False)
    .str.replace(", Not available", "", regex=False)
)
```

```
df_selected["receiver_country"] = (
    df_selected["receiver_country"]
    .str.replace("Iran, Islamic Republic of", "Iran", regex=False)
    .str.replace("Korea, Republic of", "South Korea", regex=False)
    .str.replace("Korea, Democratic People's Republic of", "North Korea", regex=False)
)

df_selected["receiver_country"] = (
    df_selected["receiver_country"]
    .str.replace("Not available, ", "", regex=False)
    .str.replace(", Not available", "", regex=False)
)
```

```
df_selected["attribution_type"] = df_selected["attribution_type"].replace({
    "Technical report (e.g., by IT-companies, Citizen Lab, EFF)":
        "Technical report",

    "Self-attribution in the course of the attack (e.g., via defacement statements on websites)":
        "Self-attribution",

    "Direct statement in media report (e.g., Reuters article cites the attribution statements by a person) / self-attribution via social media":
        "Media statement / Social media",

    "Anonymous statement in media report (e.g., Reuters article cites the attribution statements of unnamed officials, or persons with knowledge into the matter etc.)":
        "Anonymous media statement",

    "Political statement / report (e.g., on government / state agency websites)":
        "Political statement / report",

    "Attribution given, type unclear":
        "Attribution type unclear",

    "Domestic legal action":
        "Domestic legal action",

    "Media report (e.g., Reuters makes an attribution statement, without naming further sources)":
        "Media report",

    "Statement in media report and political statement/technical report":
        "Media + Political/Technical report",

    "Political statement/report and indictment / sanctions":
        "Political statement + Indictment/Sanctions",

    "Statement in media report and indictment / sanctions":
        "Media statement + Indictment/Sanctions"
})
```

```
df_selected.info()

*** <class 'pandas.core.frame.DataFrame'>
RangeIndex: 3414 entries, 0 to 3413
Data columns (total 27 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   incident_id                           3414 non-null   int64  
 1   name                                  3414 non-null   object  
 2   start_date                            3322 non-null   datetime64[ns]
 3   end_date                              1541 non-null   datetime64[ns]
 4   incident_type                         3414 non-null   object  
 5   receiver_country                      3414 non-null   object  
 6   receiver_regions                      3414 non-null   object  
 7   receiver_category                     3414 non-null   object  
 8   receiver_subcategory                  3414 non-null   object  
 9   initiator_name                        3414 non-null   object  
10   initiator_country                     3414 non-null   object  
11   initiator_category                     3414 non-null   object  
12   initiator_subcategory                  3414 non-null   object  
13   attribution_type                       3414 non-null   object  
14   attributing_actor                     3414 non-null   object  
15   cyber_conflict_issue                  3414 non-null   object  
16   zero_days                             3414 non-null   object  
17   mitre_initial_access                   3414 non-null   object  
18   mitre_impact                           3414 non-null   object  
19   has_disruption                         3414 non-null   bool    
20   data_theft                             3414 non-null   object  
21   hijacking                             3414 non-null   object  
22   impact_indicator_score                 3414 non-null   int64  
23   functional_impact                     3414 non-null   object  
24   intelligence_impact                     3414 non-null   object  
25   economic_impact_value                 3414 non-null   int64  
26   response_indicator                     3414 non-null   object  
dtypes: bool(1), datetime64[ns](2), int64(3), object(21)
memory usage: 696.9+ KB
```

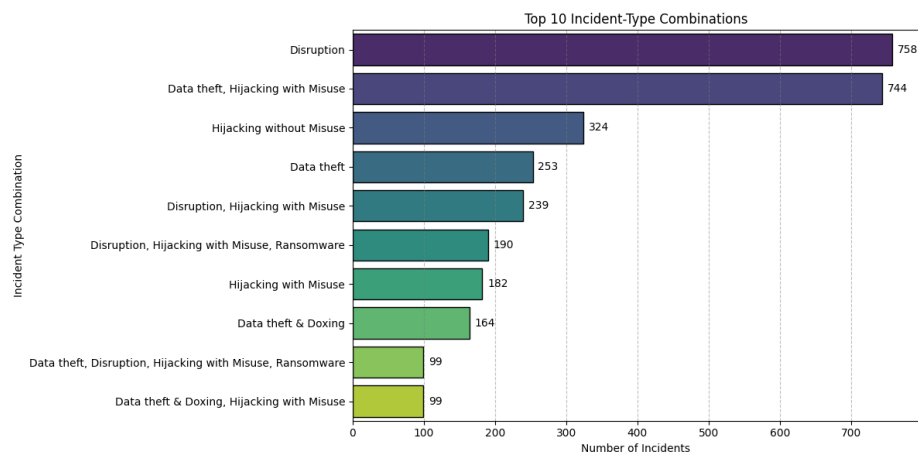
7. Data Visualization (Python)

Tools/Libraries Used:

- Python
- Pandas
- Matplotlib
- Seaborn

Visualizations Created:

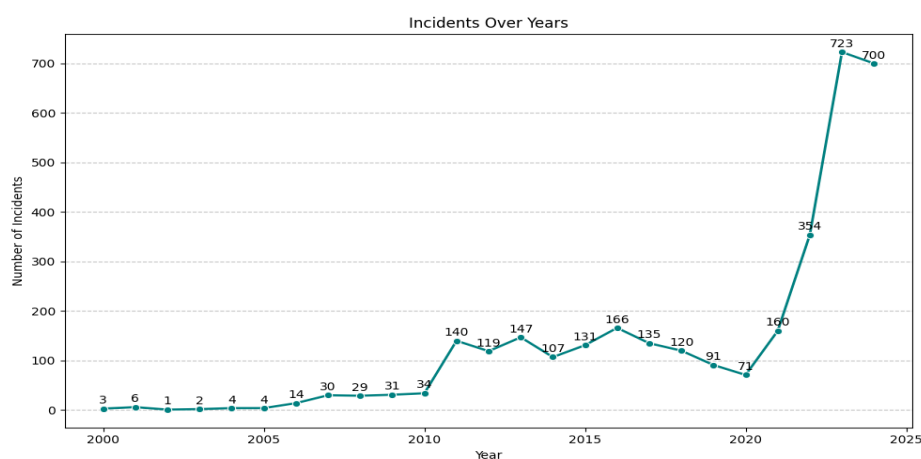
Top 10 Incident Type Combinations



Interpretation of your above chart

- **Most Common Incidents:** Disruption (758) and Data theft with Hijacking (744) are the top two attacks by a very large margin.
- **Big Drop to Third Place:** The numbers fall steeply after the top two, with 3rd place (Hijacking without Misuse) dropping all the way to 323.
- **Attacks Happen Together:** Most cases are not single actions, hackers usually combine data theft, system control, and service disruption at the same time.
- **Ransomware Is Always Linked:** Ransomware never appears alone in the top 10; it is always mixed with other attacks like disruption or hijacking (190 and 99 incidents).
- **Middle Level Threats:** Categories ranked 4 to 8 stay between 160 and 260 incidents, showing regular occurrences of data theft and doxing.
- **Main Learning for the Project:** Security teams cannot just protect against one issue; they must stop multiple combined attacks happening together.

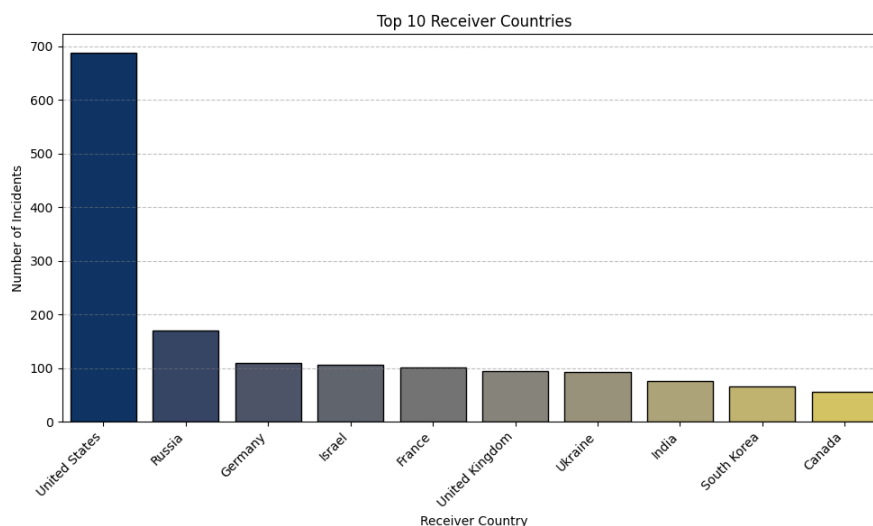
Incidents by Year



Interpretation of your above chart

- **Very Low Incidents in Early Years:** Between 2000 and 2006, reported cyber incidents were extremely low, staying under 15 cases per year.
- **First Notable Jump Around 2011:** Cases climbed from 34 in 2010 to 140 in 2011, showing the start of widespread tracking and increased internet use.
- **Steady Phase for a Decade:** From 2011 to 2019, the count stayed fairly steady between 100 and 170 incidents each year, peaking at 166 in 2016.
- **Dip During 2020:** Incidents dropped down to 71 in 2020, marking the lowest count in nearly a decade.
- **Massive Spike After 2021:** Attacks jumped dramatically from 160 in 2021 to 354 in 2022, reaching an all-time peak of 723 in 2023.
- **High Numbers Continue:** Incidents remained very high at 700 in 2024, showing that cyber threats have settled at an unprecedented high volume in recent years

Top 10 Receiver Countries

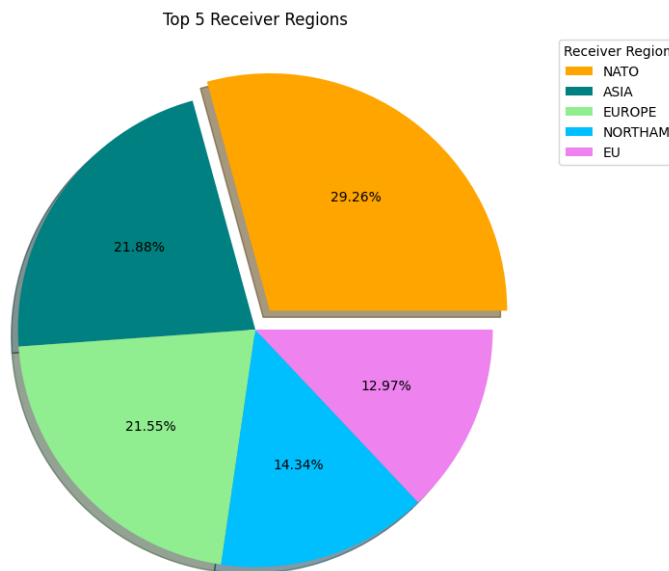


Interpretation of your above chart

- **United States Is the Top Target:** The US received the highest number of cyber incidents by far, approaching nearly 700 attacks.
- **Massive Gap Between First and Second:** Second-placed Russia has around 170 incidents, showing that the US faces almost four times more attacks than any other nation.
- **Middle Group Around 100 Incidents:** Germany, Israel, France, Ukraine, and the United Kingdom are closely grouped, each facing roughly 90 to 110 incidents.
- **Lower Tier Nations:** India, South Korea, and Canada round out the top 10, with each recording between 50 and 75 attacks.

- **Geopolitical Hotspots Dominate:** Most target countries are major global economic powers or nations involved in active geopolitical conflicts (like Ukraine, Israel, and Russia).
- **Project Takeaway:** Cyber threat actors heavily concentrate their efforts on high-value digital infrastructures, making the US and allied major economies the primary receivers of attacks.

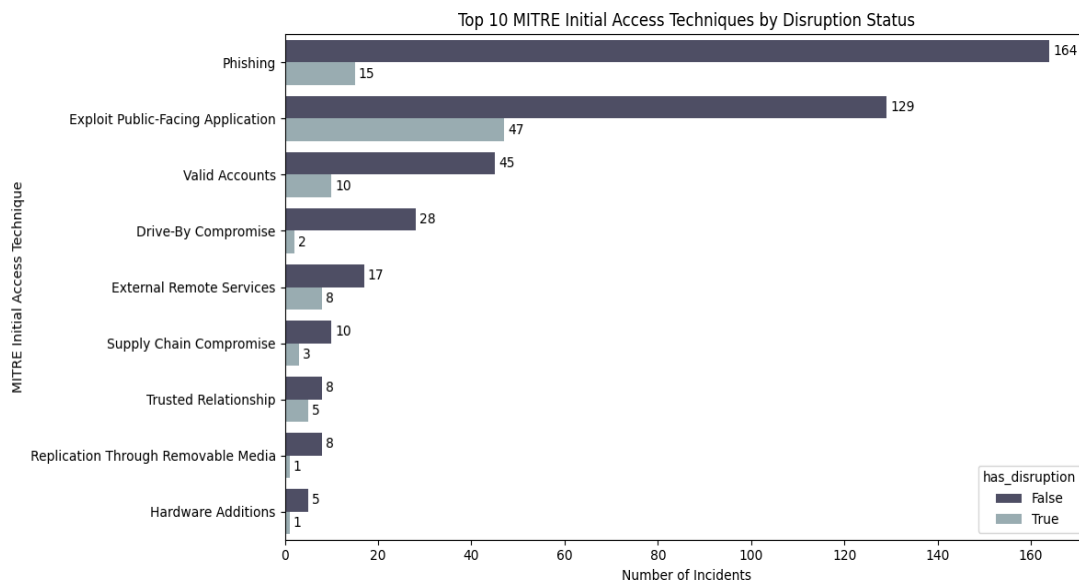
Top 5 Receiver Regions



Interpretation of your above chart

- **NATO Countries Face the Most Attacks:** The NATO alliance forms the largest share at 29.3%, highlighted with an exploded slice to show it is the primary regional target.
- **Asia and Europe Are Almost Tied:** ASIA (21.9%) and EUROPE (21.6%) hold nearly equal shares, together making up more than 43% of all attacks.
- **North America Follows in Fourth Place:** NORTHAM accounts for 14.3% of the incidents, representing a focused geographic share outside generic alliances.
- **European Union (EU) Completes the Top Five:** The EU takes the fifth spot at 13.0%, showing significant political and economic bloc targeting.
- **Western Blocs Make Up the Vast Majority:** Combined Western groupings (NATO, Europe, North America, and EU) represent over 78% of all attacks shown in the pie chart.
- **Project Takeaway:** Cyber threat operations are predominantly focused on Western strategic alliances and major Asian digital hubs, reflecting both political tensions and high-value economic targets.

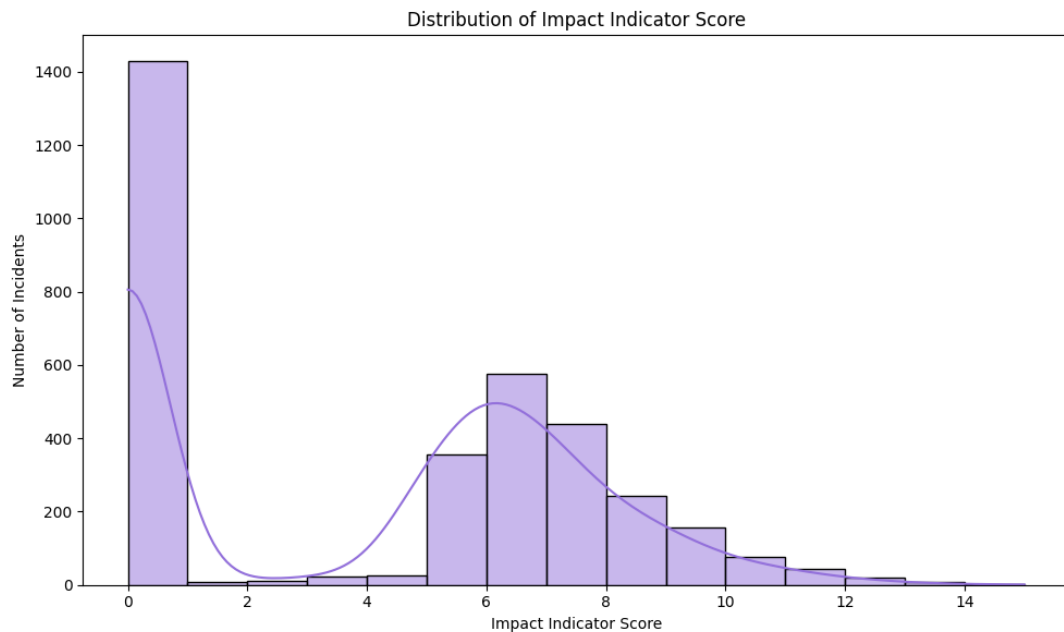
MITRE Initial Access by Disruption Status



Interpretation of your above chart

- **Non-Disruptive Incidents Dominate Overall:** Across nearly every entry vector, non-disruptive attacks (False, dark purple) heavily outnumber disruptive ones (True, light blue).
- **Phishing is Overwhelmingly Stealthy:** Phishing accounts for 164 non-disruptive incidents versus only 15 disruptive ones, showing that over 90% of phishing attacks aim for quiet objectives like credential harvesting or data exfiltration rather than taking systems down.
- **Public-Facing Application Exploits Cause the Most Disruption:** Exploit Public-Facing Application causes the highest volume of actual outages, with 47 disruptive incidents (nearly triple the disruption count of phishing).
- **Valid Accounts Follow the Stealth Trend:** Compromised accounts lead to 45 non-disruptive cases and only 10 disruptive ones, as attackers typically leverage legitimate credentials to move laterally without raising alarms.
- **Drive-By Compromise Rarely Causes Outages:** Out of 30 total incidents, 28 are non-disruptive and only 2 results in system disruption.
- **Project Takeaway:** While phishing is the most popular way to sneak inside quietly without breaking things, exploiting internet-facing applications is the leading method used when attackers actually intend to knock systems offline.

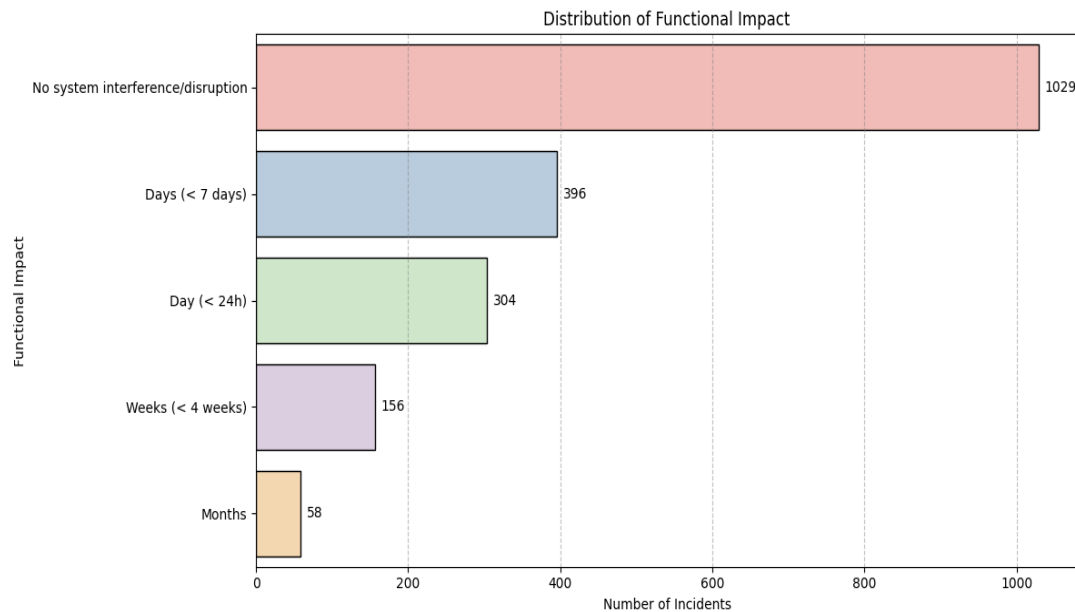
Impact Score Distribution



Interpretation of your above chart

- **Massive Peak at Zero:** Over 1,400 incidents have an Impact Indicator Score near 0, representing the single largest bar in the entire dataset.
- **Very Low Frequency from 1 to 5:** Very few incidents fall into the mild impact range between scores 1 and 5 (all well below 50 cases).
- **Clear Bell Curve for Significant Attacks:** Once attacks cause real impact, scores form a normal distribution starting at 5 and peaking between 6 and 7 (nearly 600 incidents).
- **Moderate to High Severity Cluster:** The majority of measurable-impact incidents fall within the medium-high range of scores 6 to 9.
- **Extreme Scores Are Rare:** Incidents causing severe impact with scores above 11 gradually taper off, with scores of 13 to 14 being extremely uncommon.
- **Project Takeaway:** Cyber incidents generally fall into two extremes: they either fail to register measurable impact (score 0), or they hit as structured attacks causing moderate-to-high operational damage (scores 6 to 8).

Functional Impact Distribution



Interpretation of your above chart

- **No Operational Disruption Leads:** The largest category by a huge margin is "No system interference/disruption" with 1,029 incidents, showing that most attacks do not take systems down.
- **Short-Term Downtime Dominates Disrupted Cases:** When systems actually go down, short durations are most common—Days (< 7 days) accounts for 396 incidents, followed by Day (< 24h) at 304 incidents.
- **Rapid Drop-off for Longer Outages:** Severe, prolonged outages are far less frequent, with disruptions lasting Weeks (< 4 weeks) dropping down to 156 incidents.
- **Longest Outages Are Rare:** Incidents causing system outages that last for Months are the least common, recording just 58 cases.
- **Decreasing Trend with Duration:** Across all disruption tiers, incident frequency strictly decreases as outage duration increases (from under 24 hours up to months).
- **Project Takeaway:** Cyber threats focus heavily on stealthy actions like data exfiltration without halting operations, but when operational downtime does happen, systems are usually restored within a single week.

Impact Score by Disruption Status



Interpretation of your above chart

- **Median Comparison:** The median impact score for Disrupted (True) events is slightly higher (6) than for Non-disrupted (False) events (5).
- **Distribution Spread:** Both groups show a very similar spread, with the lower 25% of cases dropping to 0 and the upper 75th percentile (Q3) hovering closely between 6 and 7.
- **Maximum Values:** Maximum scores are nearly identical (15 for False vs. 14 for True).
- **Key Takeaway:** Disruption status has minimal effect on the impact score. High-impact events occur even without disruptions, meaning disruption status alone is not a strong driver of overall impact.

8.DAX Measures & Visualizations (Power BI)

Custom DAX measures were implemented to calculate key risk metrics, high-severity events, and vulnerability exploitation counts across the dashboards:

1. High Impact Incidents:

Calculates the total number of severe cyber incidents where the overall impact severity score is 7 or higher.

File Home Help Table tools Measure tools							
Name	High impact	Home table	Cybersecurity_Cleaned_...	Format	Whole number	Uncategorized	...
1 High impact = CALCULATE(COUNTROWS(Cybersecurity_Cleaned_Data),Cybersecurity_Cleaned_Data[impact_indicator_score]>=7)							
incident_id	name	start_date	end_date	incident_type	receiver_country	receiver_regions	
1280	xHunt	01 September 2019		Hijacking without Misuse	Kuwait	ASIA, MENA, MEA, GU	
1279	Double Dragon: Crackshot backdoor (Sup	01 July 2018		Hijacking with Misuse	Southeast Asia (region)	Not available	
1275	Double Dragon: ShadowPad (Supply Chain	01 July 2017		Hijacking with Misuse	South Korea, Hong Ko	ASIA, SCS, NEA, ASIA	
1236	ZeroClear	01 January 2019		Disruption, Hijacking with Misuse	Middle East (region)	Not available	
1220	Charming Kitten Election Interference	01 July 2019		Data theft	United States, Mena R	NATO, NORTHAM, EU	
1184	Rancor Phishing	01 March 2018		Hijacking with Misuse	Southeast Asia (region)	Not available	
1092	Cobalt Dickens	01 January 2018		Data theft	United States, Australi	NATO, NORTHAM, OC	
1062	Sandvines Spyware Injection into egyptian	01 October 2017		Hijacking with Misuse	Egypt	MENA, MEA, AFRICA,	
1019	Yobit Hack by Lazarus	01 May 2017		Hijacking with Misuse	South Korea	ASIA, SCS, NEA	
976	Puppy RAT	01 January 2017		Data theft	Mena Region (region)	Not available	
837	Bellingcat-Hack 2016	01 February 2016		Data theft & Doxing, Disruption	United Kingdom	EUROPE, NATO, EU, N	
798	Tainted Leaks 2016	01 January 2016		Data theft & Doxing	United States, Russia, (	NATO, NORTHAM, EU	
793	Chinese use of zero-day Jian/EpMe	01 January 2015		Hijacking without Misuse	United States	NATO, NORTHAM	
779	Patchwork vs. Global targets related to So	01 December 2015		Data theft	United States, Global	NATO, NORTHAM	

2. Attributed Incidents:

Calculates the count of incidents successfully traced and attributed to an identified threat actor or country.

1 Attribution Count = CALCULATE(COUNTROWS(Cybersecurity_Cleaned_Data),Cybersecurity_Cleaned_Data[attribution_type] <> "Not available")							
incident_id	name	start_date	incident_type	receiver_country	receiver_regions	receiver_category	receiver_subcat
2	Azerbaijani hackers took c	01 January 2000	Disruption	Armenia	ASIA, CENTAS, CSTO	State institutions / political s	Civil service / administr
3	Armenian hacktivists targ	01 February 2000	Disruption	Azerbaijan	ASIA, CENTAS	Social groups, End user(s) / s	Advocacy / activists (
4	Chinese hacktivists target	20 May 2000	Disruption	Taiwan	ASIA, SCS	State institutions / political s	Government / ministr
5	Honker Union of China de	01 April 2001	Disruption	United States	NATO, NORTHAM	State institutions / political s	Government / ministr
6	*First Sino-US-Cyber-War*	01 May 2001	Disruption	China	ASIA, SCS, EASIA, NEA	Other	Not available
7	Textbook Hack South Kor	01 May 2001	Disruption	Japan	ASIA, SCS, NEA	Social groups, Corporate Tar	Not available
8	Prior 9/11 Taliban Hack	01 August 2001	Disruption	Afghanistan	ASIA, SASIA	Critical infrastructure	Health
9	ZeeNews/India Today Hac	22 October 2001	Disruption	India	ASIA, SASIA, SCO	Media	Not available
10	NSA vs. US muslims	01 January 2002	Data theft	United States	NATO, NORTHAM	State institutions / political s	Legislative, Civil serv
11	Titan Rain	01 January 2003	Data theft	United States, United	NATO, NORTHAM, EU	State institutions / political s	Government / ministr
13	DDoS North Korea 2004	01 April 2004	Disruption	South Korea	ASIA, SCS, NEA	State institutions / political s	Government / ministr
14	Taiwan's Kuomintang Hac	01 July 2004	Disruption	Taiwan	ASIA, SCS	State institutions / political s	Government / ministr
15	ROK Hack 2004	01 July 2004	Data theft	South Korea	ASIA, SCS, NEA	State institutions / political s	Government / ministr

Analysis and Visualizations (Power BI)

Dashboard Features & Interactivity

- To deliver actionable threat intelligence and an intuitive user experience, the report incorporates key Power BI features:
- **Interactive Slicers & Filters:** Slicers for Year, Incident Type, and Receiver Country / Initiator Country allow users to filter and slice metrics across the report.
 - **Cross-Filtering:** Selecting data points in charts (such as specific countries or incident types) automatically updates all related visuals on the page.
 - **Clear Layout & Theming:** Built with a Cyber Dark Navy theme, clear visual titles, and explicit data labels on charts for fast and easy reading.
 - **Navigation:** Used for smooth navigation across the three dashboards and quick resetting of analytical views.

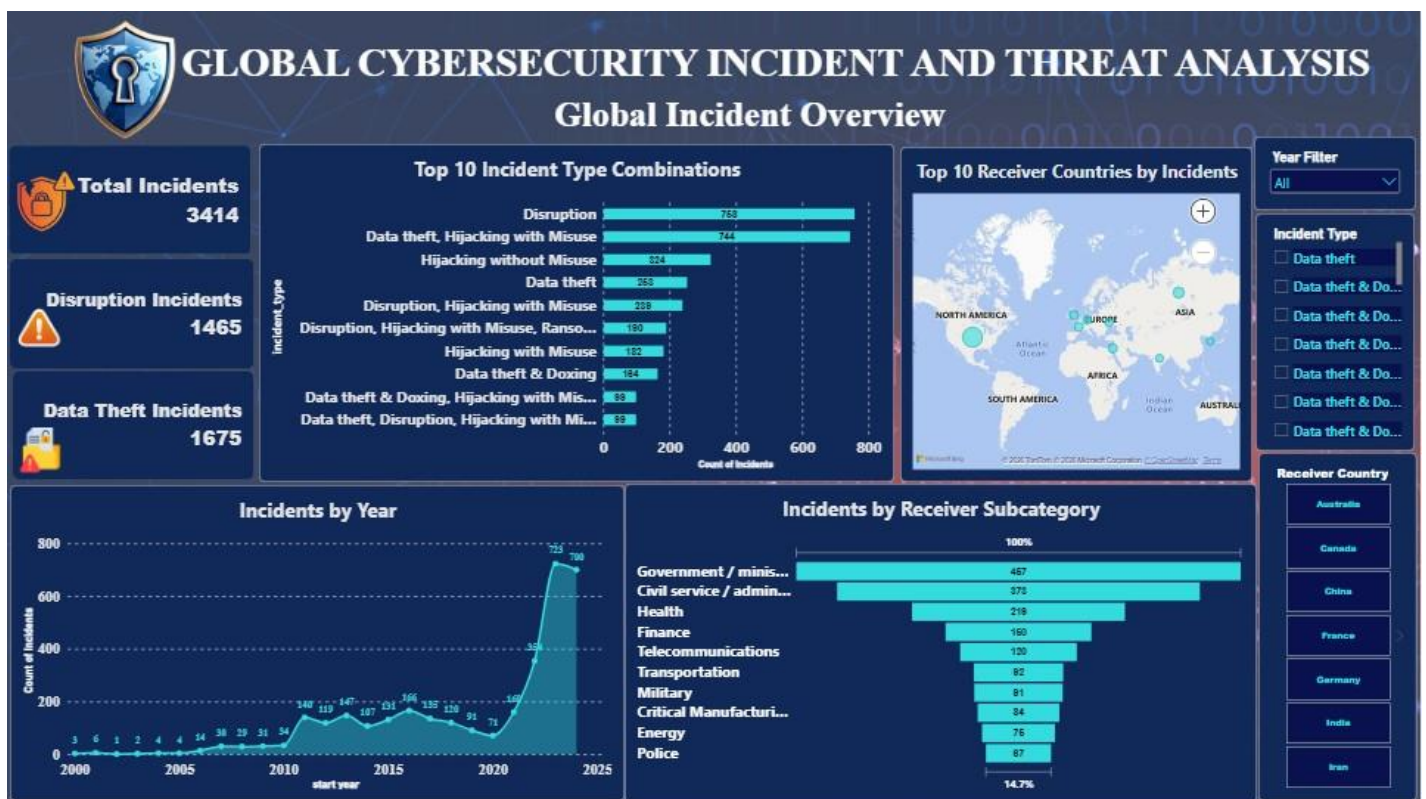
Visualizations Breakdown by Problem Statement

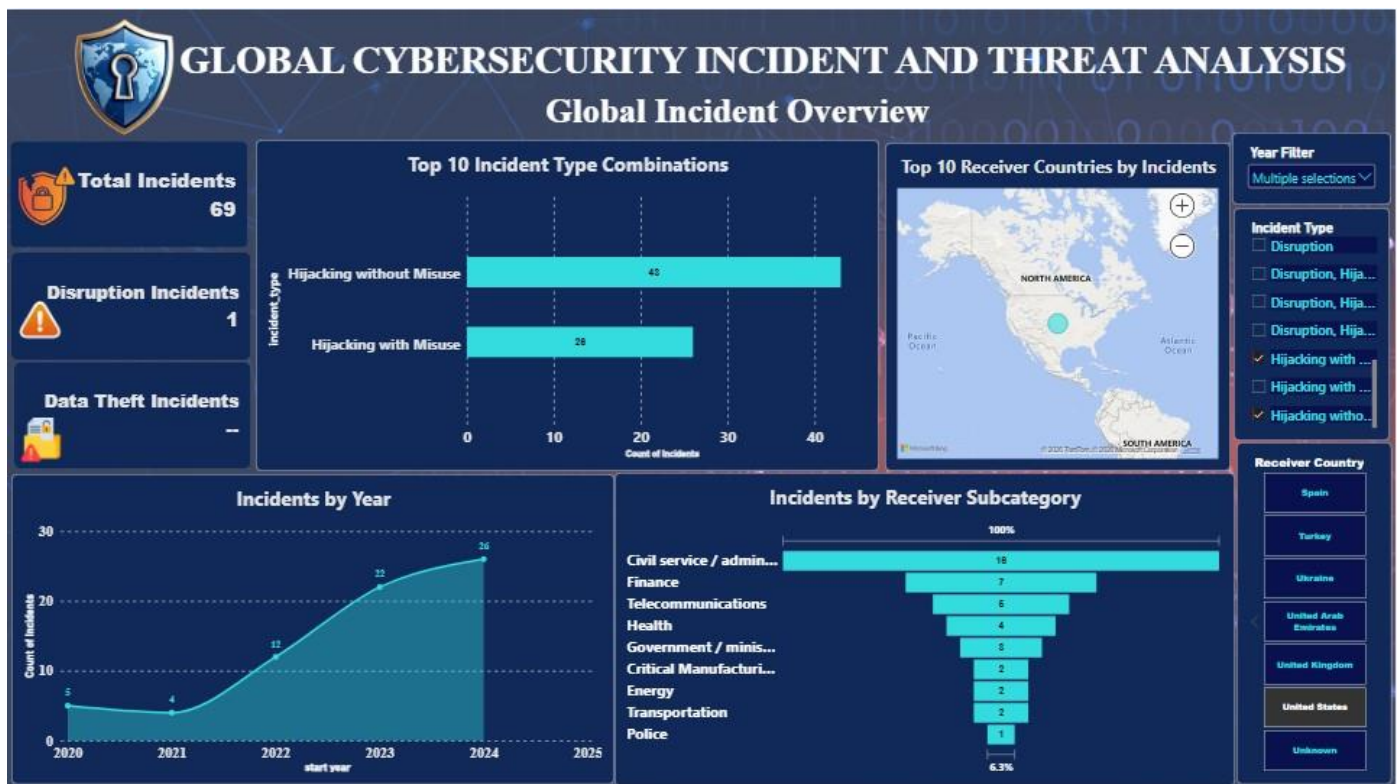
The reporting suite consists of three consolidated dashboards mapped directly to the core analytical objectives:

Dashboard 1: Global Incident Overview

- **KPI Summary Cards:** Total Incidents (3,414), Disruption Incidents (1,465), and Data Theft Incidents (1,675) providing immediate volume benchmarks.
- **Horizontal Bar Chart (Top 10 Incident Type Combinations):** Tracks top attack combinations, led by Disruption (768) and Data theft, Hijacking with Misuse (744).
- **Area / Line Trend Chart (Incidents by Year):** Illustrates longitudinal incident growth from 2000 to the post-2020 exponential surge (peaking above 700 incidents/year).
- **Map (Top 10 Receiver Countries by Incidents):** Highlights regional geographic distribution across impacted global targets.
- **Funnel (Incidents by Receiver Subcategory):** Visualises targeted public/private sectors, highlighting Government/ministries (467) and Civil service/administration (373) as top targets.

SCREENSHOT OF DASHBOARD 1: GLOBAL INCIDENT OVERVIEW

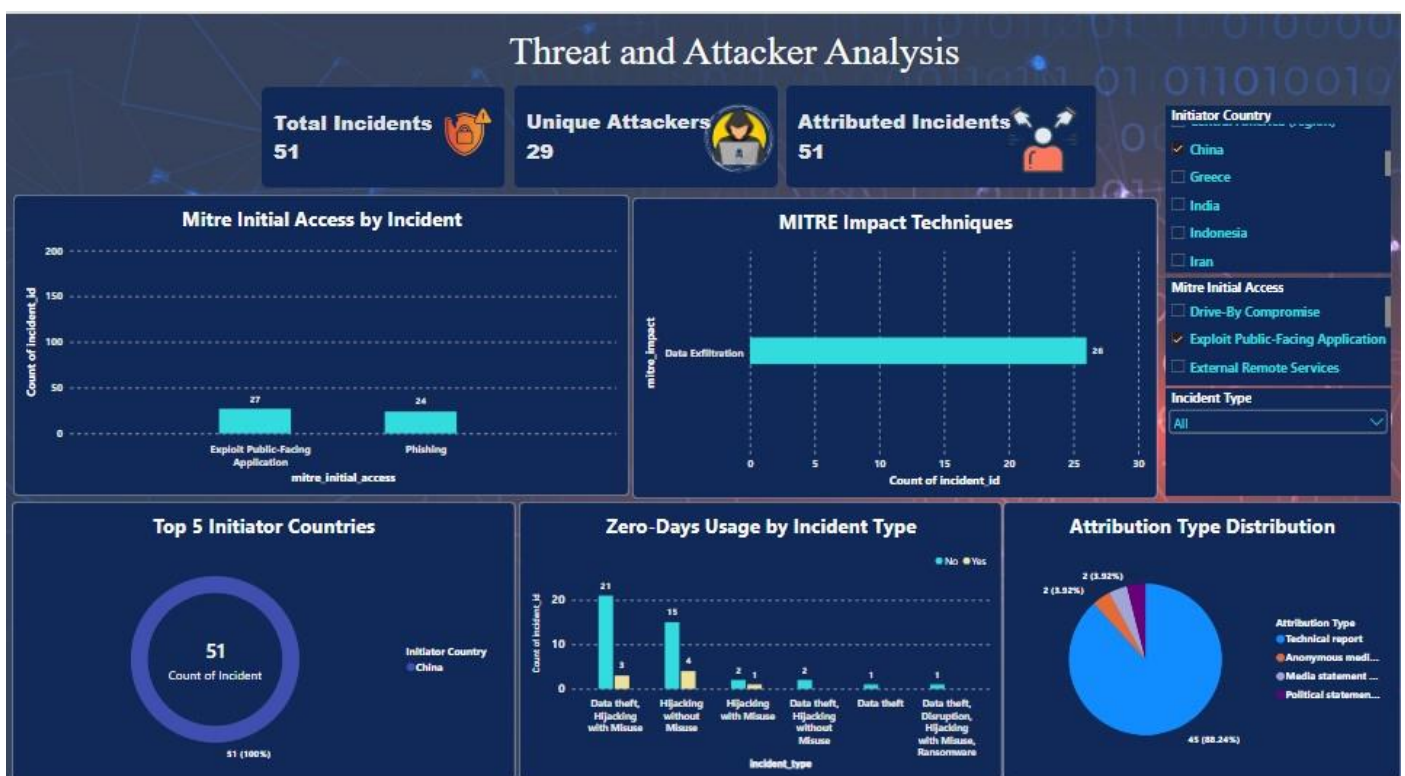
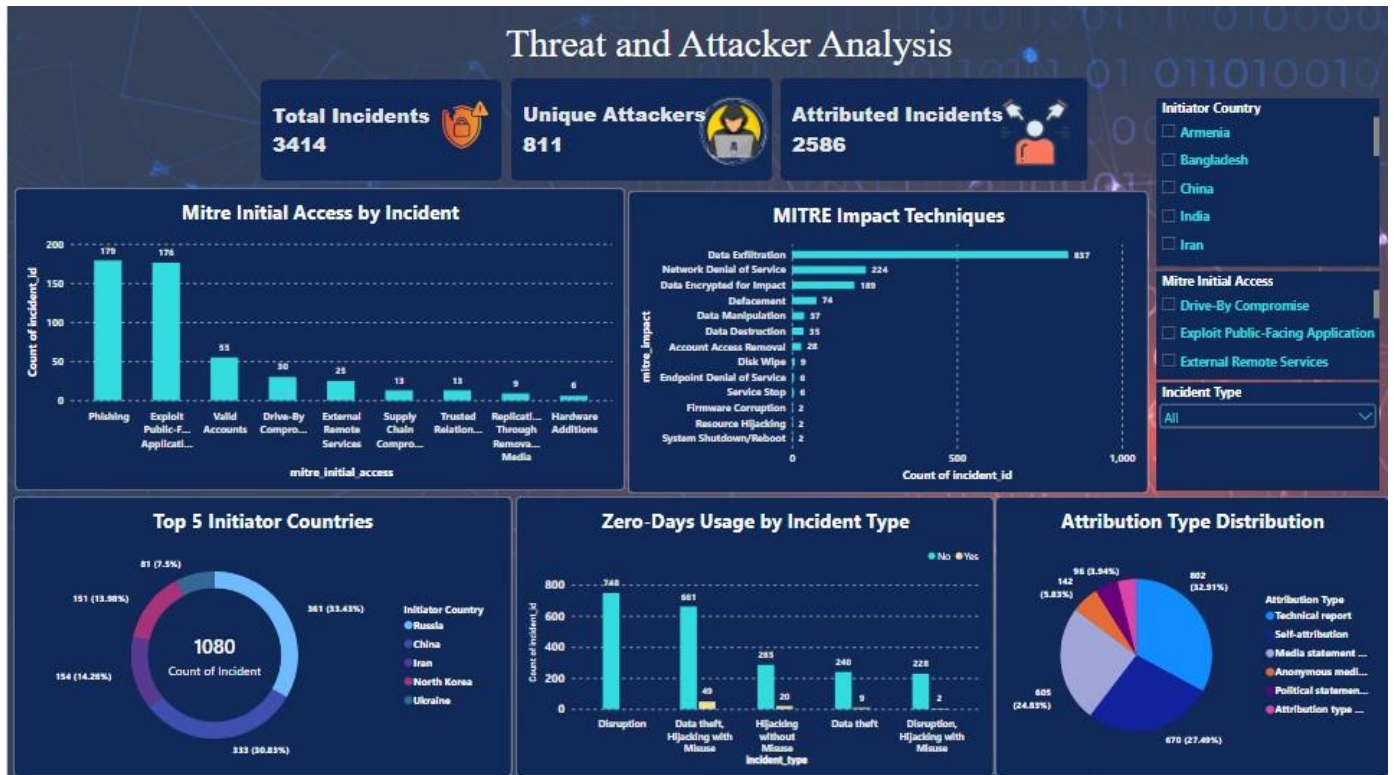




Dashboard 2: Threat and Attacker Analysis

- **KPI Summary Cards:** Highlights attribution metrics including Total Incidents (3,414), Unique Attackers (811), and Attributed Incidents (2,586).
- **Vertical Column Chart (MITRE Initial Access by Incident):** Evaluates primary entry vectors, showing heavy dominance by Phishing (179) and Exploit Public-Facing Application (176).
- **Horizontal Bar Chart (MITRE Impact Techniques):** Measures attacker impact methods, led by Data Exfiltration (837), Network Denial of Service (224), and Data Encrypted for Impact (189).
- **Donut Chart (Top 5 Initiator Countries):** Breaks down known nation-state threat group origins, led by Russia (33.43%), China (30.83%), Iran (14.26%), and North Korea (13.98%).
- **Grouped Column Chart (Zero-Days Usage by Incident Type):** Evaluates the presence of zero-day exploits across different cyber-attack categories.
- **Pie Chart (Attribution Type Distribution):** Shows confidence channels for threat attribution (Technical report at 32.91%, Self-attribution at 27.49%).

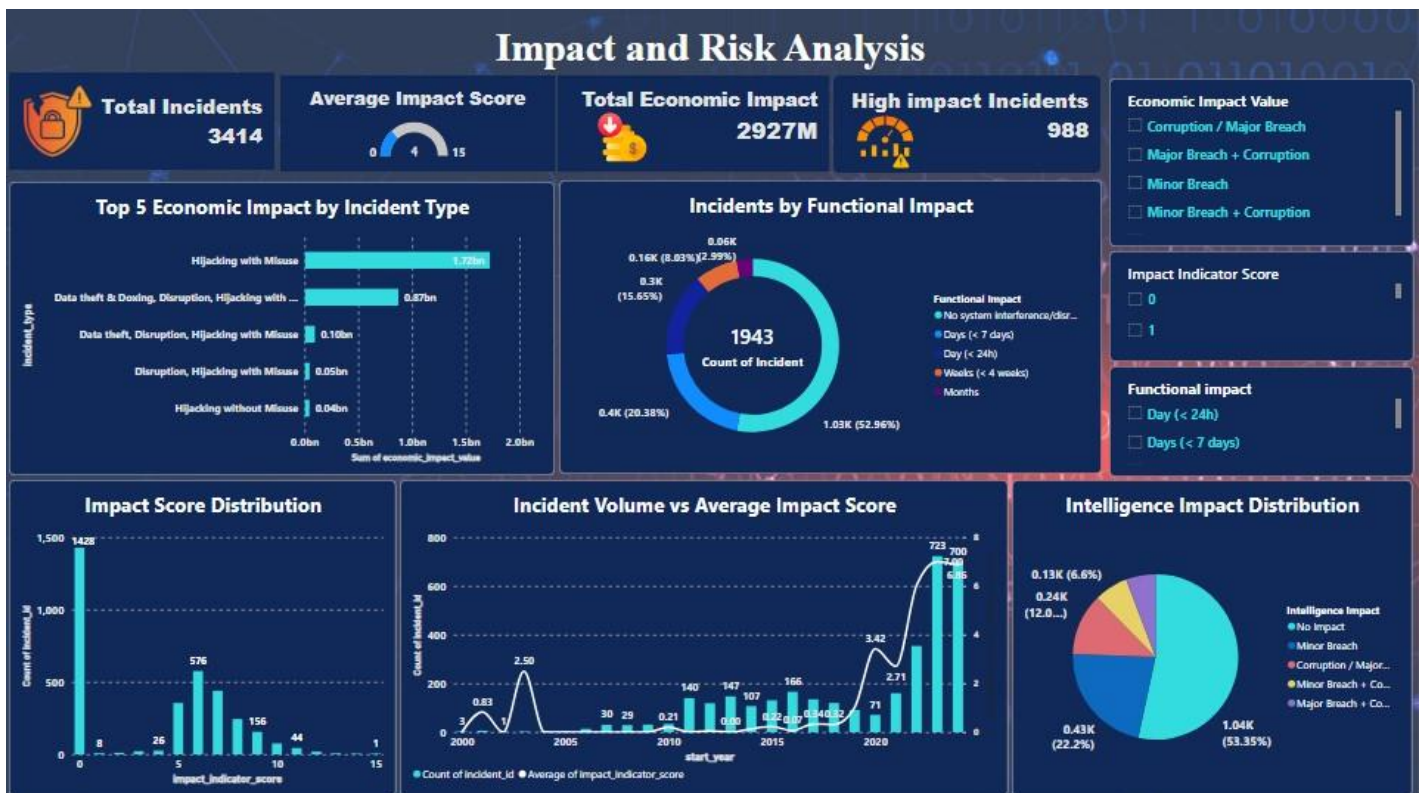
SCREENSHOT OF DASHBOARD 2: THREAT AND ATTACKER ANALYSIS

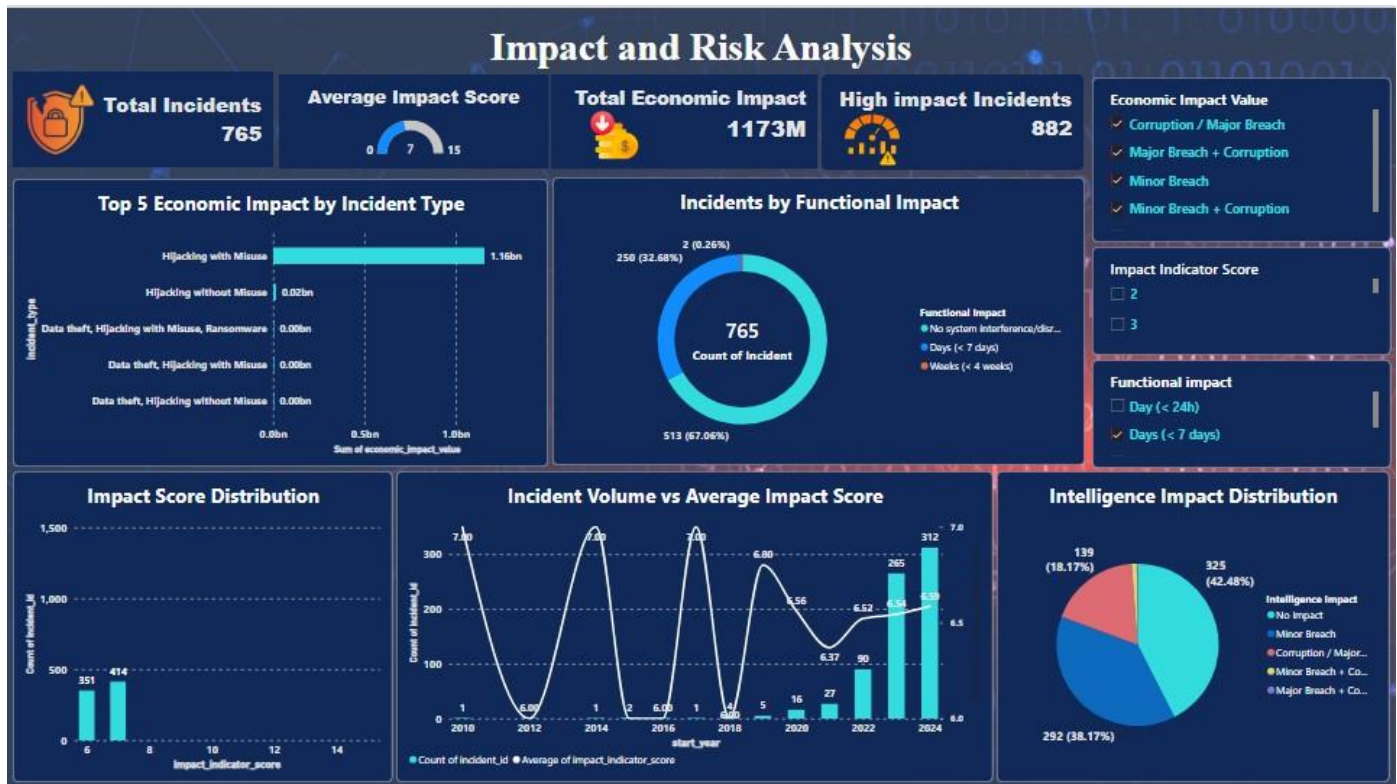


Dashboard 3: Impact and Risk Analysis

- **KPI Summary & Gauge Cards:** Total Incidents (3,414), Gauge visual for Average Impact Score (4 / 15), Total Economic Impact (\$2,927M), and High Impact Incidents (988).
- **Horizontal Bar Chart (Top 5 Economic Impact by Incident Type):** Quantifies financial losses, led overwhelmingly by Hijacking with Misuse (\$1.72B) and Data theft & Doxing (\$0.87B).
- **Donut Chart (Incidents by Functional Impact):** Classifies operational business interruption duration (e.g., No system interference at 52.96%, <24 hours at 20.38%, <7 days at 15.65%).
- **Histogram / Column Chart (Impact Score Distribution):** Shows incident severity distribution across scores 0 to 15, peaking at standard severity levels around score 5.
- **Clustered Column Chart (Incident Volume vs Average Impact Score):** Correlates incident volume growth over time against average impact severity.
- **Pie Chart (Intelligence Impact Distribution):** Tracks corporate and national intelligence compromise severity (No Impact at 53.35%, Minor Breach at 22.2%, Major Breach at 12.0%).

SCREENSHOT OF DASHBOARD 3: IMPACT AND RISK ANALYSIS





9. Insights & Conclusions

Key Findings

- **Exponential Incident Growth:** Cyber incidents maintained a stable, low baseline from 2000 to 2019, followed by an exponential surge from 2020 onward, averaging over 700 recorded incidents annually.
- **Government & Critical Infrastructure as Primary Targets:** Government ministries (467 incidents) and Civil administration (378 incidents) represent the most heavily targeted sectors, followed by Healthcare and Finance.
- **Primary Attack Vectors:** Attackers overwhelmingly gain initial access through Phishing (179) and Exploiting Public-Facing Applications (176). The most common impact goal is Data Exfiltration (837).
- **Concentrated Geopolitical Attribution:** Over 64% of attributed nation-state attacks originate from two countries: Russia (33.4%) and China (30.8%), followed by Iran (14.3%) and North Korea (14.0%).
- **Massive Economic Exposure:** Recorded incidents account for \$2.927 Billion in total economic losses, with "Hijacking with Misuse" driving the bulk of financial damages (\$1.72 Billion).

Provide the analysis insights:

Descriptive Analytics (What happened?)

- Recorded 3,414 incidents globally, dominated by data theft (1,675) and disruption (1,465).
- Traced 2,586 incidents (75.7%) to 811 unique threat actors, led by Russia and China.
- Approximately 36% of attacks caused operational downtime ranging from under 24 hours to 7 days.
- Incurred \$2.927 Billion in total economic losses, with 988 high-impact incidents.

Diagnostic Analytics (Why did it happen?)

- **Post-2020 Surge:** Rapid digital shift, cloud adoption, and geopolitical conflicts expanded target surfaces.
- **Low-Cost Entry Points:** Attackers relied on phishing and unpatched public apps rather than costly zero-day exploits.
- **High Financial Losses:** System hijacking combined with misuse caused long unauthorized access and severe business extortion.

Predictive Analytics (What is likely to happen?)

- **High Attack Volumes:** Annual cyber incidents are expected to stay above 700+ due to automated attack tools.
- **Critical Sector Targeting:** Attacks will increasingly hit Healthcare, Telecom, and Energy supply chains.
- **Double Extortion:** Attackers will continue pairing data theft with operational disruption to demand higher ransoms.

Prescriptive Analytics (What should be done?)

- **Patch Public Applications:** Secure external-facing systems quickly to block the primary entry point.
 - **Enforce MFA & Phishing Defences:** Mandate multi-factor authentication and employee security training.
 - **Adopt Zero-Trust Architecture:** Segment sensitive government and core networks to prevent data exfiltration.
 - **Maintain Offline Backups:** Build fast disaster recovery systems to keep operational downtime under 24 hours.
-

10. Conclusions

The integration of Python and Power BI proved effective for end-to-end cybersecurity data analysis, from data cleaning and preprocessing to interactive visualization and reporting. The EuRepoC Global Dataset was analysed to identify patterns in cyber incidents, threat actors, attack techniques, attribution, zero-day usage, disruption, and overall impact.

Python was used for data preprocessing, handling missing values, transforming variables, and performing exploratory data analysis. Power BI was then used to create interactive dashboards that present incident trends, geographical distribution, threat and attacker characteristics, and impact-related insights in a clear and user-friendly manner.

The analysis helped identify important patterns in cyber incidents and provided a better understanding of how different attack types, techniques, and threat actors are associated with disruption and impact. The interactive dashboards also make it easier to explore the dataset and support cybersecurity-related analysis and decision-making.

Overall, the project demonstrates how combining Python for analytical processing with Power BI for visualization can provide an effective approach for transforming a large cybersecurity dataset into meaningful and actionable insights.
